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# DETAIL SPECIFICATION

# MOUNT, RESILIENT, TYPE 5M10,000-H

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the Type 5M10,000-H resilient mount assembly consisting of the following three elements: a compression component, a shear component, and a snubber component. The mount assembly has an assigned load capacity of 5,000 to 10,000 pounds and supports loads only in its axial direction (see 6.4.1).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items

(Copies of this document are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385873 - Mount, Resilient; Type 5M10,000-H Assembly; Compression, Shear, & Snubber

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D395 - Standard Test Methods for Rubber Property – Compression Set

ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension

ASTM D429 - Standard Test Methods for Rubber Property – Adhesion to Rigid Substrates

ASTM D471 - Standard Test Method for Rubber Property – Effect of Liquids

ASTM D573 - Standard Test Method for Rubber – Deterioration in an Air Oven

ASTM D792 - Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement

ASTM D1005 - Standard Test Method for Measurement of Dry-Film Thickness of Organic Coatings Using Micrometers

ASTM D1149 - Standard Test Methods for Rubber Deterioration – Cracking in an Ozone Controlled Environment

ASTM D1229 - Standard Test Method for Rubber Property – Compression Set at Low Temperatures

ASTM D2240 - Standard Test Method for Rubber Property – Durometer Hardness

ASTM D2632 - Standard Test Method for Rubber Property – Resilience by Vertical Rebound

ASTM D5992 - Standard Guide for Dynamic Testing of Vulcanized Rubber and Rubber-Like Materials Using Vibratory Methods

ASTM D7091 - Standard Practice for Nondestructive Measurement of Dry Film Thickness of Nonmagnetic Coatings Applied to Ferrous Metals and Nonmagnetic, Nonconductive Coatings Applied to Non-Ferrous Metals



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(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

| | | |
|---|---|---|
| ISO 10846-1 | - | Acoustics and Vibration – Laboratory Measurement of Vibro-Acoustic Transfer Properties of Resilient Elements - Part 1: Principles and Guidelines |
| ISO 10846-2 | - | Acoustics and Vibration – Laboratory Measurement of Vibro-Acoustic Transfer Properties of Resilient Elements - Part 2: Direct Method for Determination of the Dynamic Stiffness of Resilient Supports for Translatory Motion |

(Copies of these documents are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The mounts furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Materials.

3.2.1 Metal. The metal components of the mount assembly shall be manufactured from steel or manganese bronze as specified (see 6.2), in accordance with Drawing 803-1385873. The metal components shall be formed to shape and shall be finished in accordance with the dimensions and allowable tolerances specified by the drawing.

3.2.2 Rubber.

3.2.2.1 Rubber elements. The components of the mount assembly shall be molded to the specified form shown on Drawing 803-1385873. The load carrying resilient elements of the shear and compression components shall be molded from a compound using natural rubber as the basic material. The natural rubber components shall be coated with an oil-ozone resistant compound. The rubber element of the snubber component shall be fabricated of an oil-resistant compound utilizing polymerized chloroprene or a copolymer product of butadiene and acrylonitrile as the basic material. Coating of the mounts may be accomplished by, but not limited to, dipping, painting, or spraying. BUNA-N components shall be coated with an oil-ozone resistant compound. Refer to 6.6 for suggested formulations for rubber stocks and coating.

3.2.2.1.1 Porosity and delamination. The rubber elements in the components of the finished mount shall have no porosity, air pockets, or delamination in any portion when visually examined. The cut specimens obtained in 4.4.13 and 4.4.19 shall also be examined for these defects.

3.2.2.1.2 Bond. The rubber elements shall be bonded to the metal components as specified on Drawing 803-1385873.

3.2.2.1.3 Surface condition. There shall be no backrinding, blisters, or any other defects on the outer surfaces of the rubber elements of the three components when examined.

3.2.2.1.4 Physical requirements.

3.2.2.1.4.1 Rubber compounds. The vulcanized rubber compounds used for the rubber elements of the shear, compression, and snubber components shall conform to the physical requirements specified in table I.



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TABLE I. Physical requirements for rubber compounds in mount components.

| Properties | Rubber elements of shear and compression components | Rubber element of snubber | Test method |
|---|---|---|---|
| Tensile strength (lb/in$^2$, min.) | | | |
| Before aging | 2,700 | 2,400 | 4.4.2 |
| After aging at 194 °F for 46 hours | 2,300 | 2,300 | 4.4.4 |
| Elongation at break (min.), percent | | | |
| Before aging | 400 | 300 | 4.4.2 |
| After aging at 194 °F for 46 hours | 400 | 250 | 4.4.4 |
| Compression set (max.) after aging at 194 °F for 46 hours, percent | 35 | --- | 4.4.3.1 |
| Cold compression set (max.), percent | 12 | --- | 4.4.3.2 |
| Volume change in oil (max.), percent | --- | 5 | 4.4.5.1 |
| Adhesion of rubber to metal (lb/in$^2$, min.) | 600 | 450 | 4.4.6.1 |
| Resilience (max.), percent | --- | 40 | 4.4.12 |
| Cut growth | No increase in length | --- | 4.4.13 |

3.2.2.1.4.2 Hardness. The hardness of the rubber compounds used for the compression, shear, and snubber components shall be determined in accordance with 4.4.10. The hardness found during qualification inspection shall be recorded as a requirement with a tolerance of ±5 points for subsequent conformance testing of the component compounds.

3.2.2.1.4.3 Specific gravity. The specific gravity of the rubber compounds used for the compression, shear, and snubber components shall be determined in accordance with 4.4.11. The specific gravity found during qualification inspection shall be recorded as a requirement with a tolerance of ±0.02 for subsequent conformance testing of the component compounds.

3.2.2.1.4.4 Oil-ozone resistant coating. The oil-ozone resistant coating shall conform to the physical requirements specified in table II (see 6.6).

TABLE II. Physical requirements for oil-ozone resistant coating.

| Property | Requirement | Test |
|---|---|---|
| Properties of dried film: | | |
| Tensile strength (lb/in$^2$, min.) | 2,000 | 4.4.2 |
| Elongation at break, percent (min.) | 400 | 4.4.2 |



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TABLE II. Physical requirements for oil-ozone resistant coating – Continued.

| Property | Requirement | Test |
|---|---|---|
| Properties of coated test specimens: | | |
| Volume change percent (max.), after immersion in oil at 158 °F for 70 hours | 5 | 4.4.5.2 |
| Adhesion of coating | | |
| Before immersion in oil | No failures | 4.4.6.2 |
| After immersion in oil at 158 °F | No failures | 4.4.6.2 |
| Ozone resistance after 1 week at 104 °F with ozone partial pressure of 100 mPa | No cracks | 4.4.7 |
| Flexibility of coating after flexing for 6 cycles to 100% elongation | No cracks | 4.4.8 |
| Properties of coated mount: | | |
| Appearance | Shall be free from blisters or other defects | table XII |
| Film thickness (min.) as measured on coated metal, inch | 0.003 | 4.4.9 |

3.2.3 Protective treatment. Unless otherwise specified (see 6.2), the minimum protective treatment shall be equivalent to that specified on Drawing 803-1385873. The metal components and elastomers used in the manufacture of the mounts shall be resistant to, or protected against, corrosion by seawater, spray, or other atmospheric conditions encountered in service.

3.2.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.5 Toxicity and prohibited materials.

3.2.5.1 Toxicity. When evaluated in accordance with 4.5, the mount shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5 and 6.10).

3.2.5.2 Prohibited materials. The mount shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.3 Requirements for finished mount.

3.3.1 Dynamic stiffness. The dynamic stiffness of individual compression and shear mounts shall be within the specified limits of table III when tested in the axial direction in accordance with 4.4.14.

TABLE III. Axial dynamic stiffness requirements for shear and compression mounts.

| Mount | Test load, pounds | Dynamic stiffness requirement, lb/in |
|---|---|---|
| Shear | 1,600 | 2,300 – 3,500 |
| Compression | 8,400 | 18,000 – 26,000 |

3.3.2 Deflection at rated load. Each shear and compression mount shall not deflect less than 0.55 inch or more than 0.78 inch when tested in accordance with 4.4.15. The mount shall not show any damage, break, or separation between the component parts.



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3.3.3 Quality of rubber-to-metal bond. Individual compression and shear mounts shall show no breaks, cracks, or tears in the rubber elements or evidence of delamination at rubber-to-metal bond interfaces when tested in accordance with 4.4.16.

3.3.4 Strength. Individual compression and shear components when tested in the axial direction in accordance with 4.4.17.1, and shear components when tested in the radial direction (see 6.4.2) in accordance with 4.4.17.2, shall not show any damage, breaks or separation in the rubber or between rubber and metal parts, or any permanent deformation in excess of 1/32 inch. The snubber components shall have axial deflections within the range of 0.39 to 0.50 inches under 50,000 pounds load and axial deflections within the range of 0.50 to 0.63 inches under 150,000 pounds load when tested in accordance with 4.4.17.3.

3.3.5 Drift. The individual compression and shear mounts shall not drift more than 50 mils under their respective loads when tested in accordance with 4.4.18. When tested after the drift test, the dynamic stiffness of the mounts shall be within the specified limits of table III.

3.3.6 Porosity and delamination. The resilient elements of the mounts shall not show evidence of porosity in the rubber elements nor separation of the rubber elements into distinct layers or laminations when tested and examined in accordance with 4.4.19.

3.3.7 Mount design. The mount shall conform to all applicable details shown on Drawing 803-1385873. Each mount component shall be identified with the markings specified on the drawing. There shall be no defects (see table XII) when examined.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be performed on rubber samples and finished mounts as specified in 4.2.1 and 4.2.2.

4.2.1 Rubber samples and tests for qualification inspection.

4.2.1.1 Rubber samples. Samples listed in table IV shall be used for the tests specified in 4.2.1.2 and 4.2.1.3.

TABLE IV. Rubber samples for qualification.

| Quantity | Type of rubber compound | Description of sample | Size of sample | Property to be measured |
|---|---|---|---|---|
| 3 | Shear and compression | Sheet | 0.08±0.010 × 6 × 6 inches | Tensile strength and ultimate elongation (initial and aged); Specific gravity (initial) |
| 3 | Snubber | Sheet | 0.08±0.010 × 6 × 6 inches | Tensile strength and ultimate elongation (initial and aged); Specific gravity (initial) |
| 3 | Shear and compression | Specimen covered with oil-ozone resistant coating | 0.08±0.010 × 1 × 2 inches | Resistance to oil, volume change |
| 3 | Shear and compression | Specimen covered with oil-ozone resistant coating | 0.08±0.010 × 1 × 6 inches | Ozone resistance, adhesion, and flexibility of coating |



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TABLE IV. Rubber samples for qualification – Continued.

| Quantity | Type of rubber compound | Description of sample | Size of sample | Property to be measured |
|---|---|---|---|---|
| 3 | Shear and compression | ASTM D429 Method A specimen | --- | Rubber-metal adhesion |
| 3 | Snubber | ASTM D429 Method A specimen | --- | Rubber-metal adhesion |
| 3 | Snubber | Sheet | 0.08±0.01 × 1 × 2 inch | Resistance to oil |
| 6 | Shear and compression | Cylinder | 0.5±0.02 inch high × 1.14±0.02 inches diameter | Compression set and hardness |
| 3 | Snubber | Cylinder | 0.5±0.02 inch high × 1.14±0.02 inches diameter | Hardness and resilience |
| 1 | Oil resistant coating | Sheet | 0.006±0.003 × 6 × 6 inches | Tensile strength and ultimate elongation |
| 1 | Shear and compression | Compression component cylinder sandwich 1/ | 1⅜ inches high × 6½ inches diameter | Cut growth |

NOTE:
1/ The cut growth sample is made from compression component cylinders referred to in 4.2.2.5.

4.2.1.2 Uncoated rubber samples. The uncoated rubber samples specified in 4.2.1.1 shall be subjected to the tests specified in table V.

TABLE V. Tests on uncoated rubber samples.

| Property | Requirement | Inspection |
|---|---|---|
| Initial tensile strength | table I | 4.4.2 |
| Initial ultimate elongation | table I | 4.4.2 |
| Tensile strength after oven aging | table I | 4.4.2 and 4.4.4 |
| Ultimate elongation after oven aging | table I | 4.4.2 and 4.4.4 |
| Compression set after oven aging (shear/compression stock only) | table I | 4.4.3.1 |
| Cold compression set (shear/compression stock only) | table I | 4.4.3.2 |
| Hardness | 3.2.2.1.4.2 | 4.4.10 |
| Specific gravity | 3.2.2.1.4.3 | 4.4.11 |
| Resistance to oil (snubber stock only) | table I | 4.4.5.1 |
| Adhesion to metal | table I | 4.4.6.1 |
| Resilience (snubber stock only) | table I | 4.4.12 |
| Cut growth (shear/compression stock only) | table I | 4.4.13 |



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4.2.1.3 Oil-ozone resistant coating samples. The oil-ozone resistant coating and coated rubber samples specified in 4.2.1.1 shall be subjected to the tests specified in table VI.

TABLE VI. Tests on coated rubber and dry film samples.

| Property | Requirement | Inspection |
|---|---|---|
| Tensile strength of dried film | table II | 4.4.2 |
| Ultimate elongation of dried film | table II | 4.4.2 |
| Volume change in oil | table II | 4.4.5.2 |
| Adhesion of coating | table II | 4.4.6.2 |
| Ozone resistance | table II | 4.4.7 |
| Flexibility of coating | table II | 4.4.8 |

4.2.2 Qualification inspection on finished mounts.

4.2.2.1 Mount samples. If the rubber samples listed in table IV meet all requirements in 4.2.1, the tests specified in 4.2.2.2 through 4.2.2.5 shall be conducted using four compression components, four shear components, two snubber components, and one additional compression component (table IV, cut growth) manufactured from the compounds represented by the rubber samples.

4.2.2.2 Group one. Two compression components and two shear components specified in 4.2.2.1 shall be subjected to the tests specified in table VII:

TABLE VII. Test group one.

| Test | Requirement | Inspection |
|---|---|---|
| Examination | No defects | table XII |
| Dynamic stiffness | 3.3.1 | 4.4.14 |
| Deflection at rated load | 3.3.2 | 4.4.15 |
| Quality of rubber-to-metal bond | 3.3.3 | 4.4.16 |
| Strength, axial (compression/shear mounts) | 3.3.4 | 4.4.17.1 |
| Strength, radial (shear mounts only) | 3.3.4 | 4.4.17.2 |
| Film thickness of oil-ozone resistant coating | table II | 4.4.9 |



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4.2.2.3 Group two. The remaining two compression components and two shear components specified in 4.2.2.1 shall be subjected to the tests specified in table VIII.

TABLE VIII. Test group two.

| Test | Requirement | Inspection |
|---|---|---|
| Examination | No defects | table XII |
| Dynamic stiffness | 3.3.1 | 4.4.14 |
| Deflection at rated load | 3.3.2 | 4.4.15 |
| Quality of rubber-to-metal bond | 3.3.3 | 4.4.16 |
| Drift | 3.3.5 | 4.4.18 |
| Porosity and delamination | 3.3.6 | 4.4.19 |

4.2.2.4 Snubber. The two snubber samples specified in 4.2.2.1 shall be subjected to the strength test in accordance with 4.4.17.3.

4.2.2.5 Compression component. The additional compression component specified in 4.2.2.1 shall be used in preparing specimens for conducting the cut growth test specified in 4.4.13.

4.3 Conformance inspection. Conformance inspection shall be performed on each production lot and include tests and examinations contained in 4.3.2 and 4.3.3.

4.3.1 Lot. For the purpose of conformance inspection and sampling, a lot shall consist of all the mounts of one type, design, and load rating offered for delivery. A lot serial number shall be assigned to each mount. The lot serial number shall not be repeated in any one quarter. The serial number shall be traceable to the mount and snubber rubber batch numbers, the manufacturing and process records against which the mounts and snubbers were manufactured, and all conformance requirements as required by this specification.

4.3.2 Rubber samples and tests for conformance inspection.

4.3.2.1 Rubber compound samples. Samples, as specified in table IX, shall be provided from each batch of shear and compression component stock and snubber component stock, which is mixed for manufacture of the mount components in the lot. The samples shall be certified to be of the same material and equivalent cure as the corresponding components in the lot of finished mounts offered for delivery.

TABLE IX. Rubber samples for conformance.

| Quantity | Type of rubber compound | Description of sample | Size of sample | Property to be measured |
|---|---|---|---|---|
| 2 | Shear and compression | Sheet | 0.08±0.010 × 6 × 6 inches | Tensile strength, ultimate elongation, and specific gravity |
| 2 | Snubber | Sheet | 0.08±0.010 × 6 × 6 inches | Tensile strength, ultimate elongation, and specific gravity |
| 1 | Shear and compression | Slab or disc | 0.5 inch thick | Hardness, compression set |
| 1 | Snubber | Slab or disc | 0.5 inch thick | Hardness |



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4.3.2.2 Conformance tests on rubber samples. Samples as specified in 4.3.2.1 shall be subjected to the tests specified in table X:

TABLE X. Conformance tests on rubber samples.

| Test | Requirement | Inspection |
|---|---|---|
| Initial tensile strength | table I | 4.4.2 |
| Initial ultimate elongation | table I | 4.4.2 |
| Compression set | table I | 4.4.3.1 |
| Hardness | 3.2.2.1.4.2 | 4.4.10 |
| Specific gravity | 3.2.2.1.4.3 | 4.4.11 |

4.3.2.3 Rejection. If any samples tested in accordance with 4.3.2.2 fail to meet the requirements, all mounts in the lot represented by the samples shall be rejected.

4.3.3 Conformance inspection on mounts.

4.3.3.1 Visual and dimensional examination.

4.3.3.1.1 Sampling for visual and dimensional examination. As a minimum, a random sample of mounts consisting of an individual compression, shear, and snubber components shall be taken in accordance with table XI from each lot for visual and dimensional examination. Classifications and types of physical defects are shown in table XII. MIL-STD-407 shall be used to determine and evaluate defects.

4.3.3.1.2 Rejection for visual and dimensional examination. If one or more defects are found in any sample, the defective components shall be rejected and the entire lot screened for all defects. All defective components shall be rejected and removed from the lot.

TABLE XI. Sampling for visual and dimensional examination.

| Number of mounts in lot | Number of mounts in sample |
|---|---|
| 90 and under | 8 |
| 91 to 200 | 20 |
| 201 to 300 | 30 |

TABLE XII. Classification of physical defects.

| Category | Defects |
|---|---|
| *Major:* | |
| 101 | Evidence of use of unauthorized materials. |
| 102 | Rubber elements not molded to specified form on drawing. |
| 103 | Evidence of delamination or air pockets in rubber elements of finished product. |
| 104 | Rubber elements not bonded to metal components in conformance with drawing. |
| 105 | Evidence of backrinding, blisters, cracks, breaks, gouges, tears, or other imperfections on outer surface of rubber elements. |
| 106 | Evidence of tackiness, non-drying, peeling, or other imperfections in the oil-resistant coating or the required coating is absent. |



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TABLE XII. Classification of physical defects – Continued.

| Category | Defects |
|---|---|
| *Major:* | |
| 107 | Metal components not protected from corrosion by seawater, spray, or other atmospheric conditions encountered in service. |
| 108 | Dimensions, configuration, and construction not in accordance with drawing. |
| 109 | Any other defect that would affect the serviceability, performance, or service life of the mounts. |
| *Minor:* | |
| 201 | Identification marking not distinct. |
| 202 | Burrs, rough edges, and sharp corners not removed. |
| 203 | Any defect that would not affect the serviceability, performance, or service life of the mounts. |

4.3.3.2 Conformance tests on mounts.

4.3.3.2.1 Conformance test to be performed on all mounts in the lot. Prior to sampling for conformance tests, all mounts in the lot shall be tested for deflection at rated load in accordance with 4.4.15. All mounts not meeting the requirements of 3.3.2 shall be rejected and removed from the lot.

4.3.3.2.2 Sampling for tests. A random sample of mounts shall be taken from each lot in accordance with table XIII for the tests specified in 4.3.3.2.3.

TABLE XIII. Sampling for conformance tests.

| Number of mounts in a lot | Number of mounts in a sample |
|---|---|
| 3 to 15 | 3 |
| 16 to 20 | 4 |
| 21 to 25 | 5 |
| 26 to 90 | 8 |
| 91 to 150 | 13 |
| 151 to 280 | 20 |
| 281 to 300 | 32 |

4.3.3.2.3 Tests to be performed on mount samples. Each of the mounts selected in accordance with 4.3.3.2.2 shall be subjected to the tests specified in table XIV.

TABLE XIV. Conformance tests on mount samples.

| Test (listed in sequence of performance) | Mount component(s) | Requirement | Inspection |
|---|---|---|---|
| Thickness of oil-ozone resistant film | Compression | table II | 4.4.9 |
| Dynamic stiffness | Shear and compression | 3.3.1 | 4.4.14 |
| Quality of rubber-to-metal bond | Shear and compression | 3.3.3 | 4.4.16 |



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4.3.3.2.4 Rejection for mount conformance tests. If the quality of rubber-to-metal bond requirements have not been met by all mount components in the sample, the entire lot shall be rejected. If dynamic stiffness and thickness of oil-ozone resistant film requirements have not been met by all components in the sample, the defective component(s) shall be rejected and the entire lot shall be tested for the defect(s) found. All defective components shall be rejected and removed from the lot.

4.4 Method of tests.

4.4.1 Standard test conditions. Unless otherwise specified in the test procedure, tests shall be made within an ambient temperature range of 65 to 80 degrees Fahrenheit (°F).

4.4.2 Tensile and ultimate elongation tests. Tensile strength and ultimate elongation tests shall be conducted on the compression, shear, and snubber compounds and the 0.006-inch thick coating film in accordance with ASTM D412. Die C test specimens shall be used.

4.4.3 Compression set.

4.4.3.1 Compression set after aging. Compression set shall be determined in accordance with ASTM D395 Method B, except oven aging conditions shall be 194±2 °F for 46±¼ hours.

4.4.3.2 Cold compression set. Cold compression set shall be determined in accordance with ASTM D1229, except the exposure shall be at 30±2 °F for 94±½ hours. Compression set shall be determined 30 minutes after release from compression.

4.4.4 Oven aging test. Specimens for tensile, ultimate elongation, and compression set tests shall be oven aged in accordance with ASTM D573 at a temperature of 194±2 °F for 46±¼ hours. Final determination of aged tensile and ultimate elongation specimens shall be made between 16 and 48 after removal from the oven. Tensile and ultimate elongation tests on unaged specimens shall be made immediately prior to and on the same machine as the tensile tests on the oven aged specimens (see 4.4.2).

4.4.5 Swelling in oil.

4.4.5.1 Volume change in oil (snubber compound). Volume change shall be determined in accordance with ASTM D471. The specimen shall be immersed in IRM 903 reference oil in accordance with ASTM D471. The immersion period shall be 46±¼ hours at 73 °F.

4.4.5.2 Volume change in oil (coated sample). Volume change in oil shall be determined in accordance with ASTM D471, except the three coated 0.08- by 1- by 2-inch specimens shall be as described in table IV and the immersion period shall be 70±¼ hours at 158±2 °F. The specimen shall be immersed in IRM 903 reference oil in accordance with ASTM D471.

4.4.6 Adhesion tests.

4.4.6.1 Adhesion to metal tests. The adhesion tests shall be in accordance with Method A of ASTM D429. Three specimens of each rubber compound (compression, shear, and snubber) shall be tested and the results averaged for each compound.

4.4.6.2 Adhesion of coating (coated sample). The adhesion of the oil-resistant coating to three 0.08- by 1- by 2-inch samples of the compression and shear rubber stock (see table IV) shall be determined before and after immersion in oil in accordance with 4.4.5.2. Each coated specimen shall be flexed, elongated by hand, and then visually examined for adhesion failures. The coated specimens shall not exhibit cracks, breaks, tears, or blisters conducive to peeling of the coating by hand either before or after immersion in oil.



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4.4.7 Ozone resistance. The ozone resistance of the coated resilient components of the mounts shall be determined on two coated 1- by 6- by 0.080-inch thick specimens (see table IV). The specimens shall be elongated to 20 percent extension. A thin layer of melted paraffin wax shall be applied to each of the four surfaces of the stretched specimens in an area not exceeding a ¼-inch width from the clamps. The stretched specimens shall be conditioned for 16±2 hours at 104±2 °F before being exposed, in accordance with ASTM D1149. The standard ozone partial pressure shall be 100 millipascals (mPA) ±10 percent, the temperature shall be 104±2 °F, and the period of exposure shall be 168 hours. Observation magnification shall be 7X.

4.4.8 Flexibility of the coating. The flexibility of the coated 1- by 6- by 0.080-inch resilient element rubber specimens specified in table IV shall be determined. Each specimen shall be clamped in a universal test machine and a tensile force applied to elongate the specimen 100 percent at a loading rate of 20 inches per minute. The force shall then be released so that the specimen is no longer elongated. The cycle shall be repeated six times. The coated specimens shall be visually examined and shall exhibit no cracks, breaks, tears, or blisters.

4.4.9 Film thickness of the oil resistant coating. The film thickness of the oil resistant coating shall be measured on the coated steel plates of the compression component using a micrometer for conformance with the requirement in table II. The thickness of the steel plate shall be determined before and after coating. The difference in the two thickness measurements divided by two shall be the film thickness of the oil resistant coating. Refer to ASTM D1005. In lieu of the above procedure, the film thickness may also be measured by using a magnetic type thickness gage in accordance with ASTM D7091.

4.4.10 Hardness. The hardness of the 0.50-inch thick specimens representing the rubber components of the mounts and snubber shall be determined in accordance with ASTM D2240. A 3-second reading shall be taken to determine conformance with 3.2.2.1.4.2.

4.4.11 Specific gravity. The specific gravity of the rubber compounds used for the compression, shear, and snubber components shall be determined on specimens cut from 0.08- by 6- by 6-inch sheets using ASTM D792. The average of three determinations shall be obtained to determine conformance with 3.2.2.1.4.3.

4.4.12 Resilience test. The resilience of the 0.50-inch thick specimens representing the rubber component of the snubber shall be determined in accordance with ASTM D2632. Three specimens shall be tested and the results averaged.

4.4.13 Cut-growth test.

4.4.13.1 Test specimen. The test specimen shall be one of the three rubber steel sandwiches of the compression component, consisting of the rubber element and the two steel plates to which it is bonded.

4.4.13.2 Apparatus. The test apparatus shall consist of two 10- by 10- by 1-inch steel plates bolted together through steel tube spacers as shown on figure 1. The length of the spacers shall be equal to 80 percent of the thickness of the rubber element of the specimen plus the thickness of the two mount plates.

4.4.13.3 Procedure.

4.4.13.3.1 Preparation of specimen. The test specimen shall be inserted between the plates of the test apparatus, and the bolts tightened to pull the steel plates against the spacers. Six horizontal cuts, about ¼ inch long and ⅛ inch deep, shall be made at 60 degree intervals around the rubber element, parallel to the mount plates and midway between them as shown on figure 1. A line shall be drawn at the end of each cut and perpendicular to it so that any growth of the cut during the test will be apparent.

4.4.13.3.2 Oven curing of specimen. The test assembly shall be placed in an oven maintained at 200±2 °F. The test assembly shall be removed from the oven after 16 hours, and the cuts examined for increase in length. There shall be no increase in cut length when examined.



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4.4.14 Dynamic stiffness test. Dynamic stiffness shall be measured individually for each compression mount and shear mount in the axial direction at test loads of 8,400 and 1,600 pounds, respectively, to determine conformance with 3.3.1. The double amplitude of vibration for each mount shall be 0.020 inch peak-to-peak at 5 Hertz (Hz). Prior to conducting dynamic stiffness tests, all mounts should first be "conditioned" a minimum of three cycles to the test load. General guidelines and considerations for dynamic stiffness testing given in ASTM D5992, ISO 10846-1, and ISO 10846-2 shall be followed.

4.4.14.1 Test system for measuring dynamic stiffness. The test system shall be capable of superimposing cyclical vibration while statically loading the mount. Refer to ASTM D5992 for examples of non-resonant forced vibration test systems and guidance in addressing mechanical and instrumentation factors that influence dynamic measurements. A commonly used type of equipment for non-resonant testing is a closed-loop servo-hydraulic machine (see figure 2). Transducers shall be arranged to measure force at the blocked output side of the mount (foundation) and displacement at the input side of the mount (equipment). The stiffness of the test apparatus (machine, load-frame, force transducer, etc.) in series with the mount under test shall be determined experimentally (see 6.9) or via calculation. A test apparatus which is at least 100 times stiffer than the dynamic stiffness of the mount being tested shall not require measurement adjustment to account for the flexibility of the test apparatus. If the test apparatus fails to meet this relative stiffness requirement, the test apparatus compliance shall be subtracted from the measurement to obtain the dynamic stiffness of the mount ($K_{mount}$). The dynamic stiffness of the mount can be calculated using the classic equation governing the summation of springs in series:

$$(1/K_{measured}) = (1/K_{mount}) + (1/K_{apparatus}) \qquad \text{yielding} \qquad K_{mount} = \frac{(K_{apparatus})(K_{measured})}{(K_{apparatus}) - (K_{measured})}$$

To avoid over-correction, the stiffness of the test apparatus shall be greater than 30 times the stiffness of the mount.

4.4.14.2 Confirmation of the measurement system. Refer to ASTM D5992 for mechanical and instrumentation factors influencing dynamic measurement, selection, and use of springs to assess the capability of the test apparatus to measure dynamic stiffness. Prior to conducting dynamic stiffness measurements on mounts, the dynamic calibration of the complete system shall be confirmed using steel springs of known spring constants having negligible damping. Springs shall be free of hysteresis and their dynamic and static stiffness (tangent to the slope of the load-deflection curve) equivalent. This procedure is not intended to replace the periodic calibration of transducers, signal conditioners, and other measuring instruments. Springs shall be selected on the basis of their linear force-deflection properties and ability to produce a spring constant comparable to the stiffness of the mount to be tested. Refer to ASTM D5992 for factors influencing the selection of springs. System confirmation shall be performed using springs of known spring constants at the excitation amplitudes and frequencies at which the mount is to be tested. System confirmation shall be performed at loads that produce spring constant measurements that bracket the expected dynamic stiffness range of the mount to be tested. This shall be accomplished by adding or removing springs within the stiffness range of intended mount measurements. Care shall be taken to ensure measurements are conducted within the linear load range of the springs. All spring dynamic stiffness measurements (corrected for test apparatus stiffness) should be within 2 percent of the known spring constants. If this is not the case, the error shall be investigated and corrected. A system scaling factor may be applied to measured dynamic stiffness with NAVSEA approval.

4.4.14.3 Measuring mount dynamic stiffness. Dynamic stiffness shall be measured individually for each compression mount and shear mount in the axial direction at loads of 8,400 and 1,600 pounds, respectively, to determine conformance with 3.3.1. Sinusoidal excitation displacement amplitude shall be approximately 0.020 inch peak-to-peak at 5 Hz for all dynamic stiffness testing. Dynamic stiffness shall be measured at the excitation frequency by using Fast Fourier Transform (FFT) based methods. This requires measurement of the sinusoidal displacement at the input of the mount and transmitted dynamic force at the blocked output of the mount, as well as the phase relationship between them. Dynamic stiffness is calculated from the measurements as follows:



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$K_{measured} = (F/X)(\cos(\theta))$

Where (F) and (X) are the amplitudes of sinusoidal force and displacement, respectively, and ($\theta$) is the phase between them. Loss factor ($\eta$) can be calculated from the measurements as follows:

$\eta = \tan(\theta)$

In order to obtain mount stiffness ($K_{mount}$) to determine compliance with 3.3.1, the measured stiffness ($K_{measured}$) may likely require adjustment to account for test apparatus flexibility (see 4.4.14.1).

$$K_{mount} = \frac{(K_{apparatus})(K_{measured})}{(K_{apparatus}) - (K_{measured})}$$

A system scaling or multiplication factor (C) applied to the measured dynamic stiffness ($C*K_{measured}$) is assumed to be unity (C=1) and not shown in the above equation.

4.4.15 Test for deflection at rated load. A universal type testing machine shall be used. Each shear component and each compression component shall be tested separately. Each component shall be subjected to a single loading cycle in the axial direction. The loadings shall be accomplished at a constant rate such that the maximum load is attained in not less than 1 minute. The deflection at rated load shall be measured to the closest 0.001 inch. Data shall be corrected for error resulting from flexibility in the test apparatus. The rated loads are 8,400 pounds for the compression component and 1,600 pounds for the shear component. The deflection of the upper rated load shall be recorded to determine conformance with 3.3.2.

4.4.16 Quality of the rubber-to-metal bond. The load on the mounts tested, as specified in 4.4.15, shall be increased to 2 times the upper rated load after recording the deflection at the upper rated load. The rate of loading should be such that the maximum load is attained in not less than 2 minutes. The mount shall be held at 2 times its upper rated load while the appearance of the resilient element and of the rubber to metal bond is examined to determine conformance to 3.3.3.

4.4.17 Strength tests. A universal type testing machine shall be employed to conduct strength tests. The testing machine shall load the components at a constant rate of deflection. Data shall be corrected for error resulting from flexibility in the test apparatus. Each component shall be examined during and after each test. Components shall not show any damage, breaks or separation in the rubber or between rubber and metal parts, or any appreciable permanent deformation.

4.4.17.1 Tests on shear and compression components in the axial direction. Each shear and compression mount shall be individually subjected to four loading and unloading cycles in the axial direction to 3 times its respective rated load. The loadings shall be conducted at a rate not to exceed 0.5 inch per minute. The deflection of each respective rated load shall be noted (see 4.4.17.2) during the fourth cycle.

4.4.17.2 Test on the shear components in the radial direction. Load deflection tests in the radial direction shall be conducted only on the shear components in an assembly similar to that shown on figure 3. Each shear component shall be compressed axially to a deflection corresponding to its rated load (see 4.4.17.1). This compression is obtained by adjusting the nuts on the four bolts, which changes the distance between the brackets and plane of the loading plate. After assembly, the shear components shall be loaded radially through four loading and unloading cycles to 30,000 pounds. The four cycles shall be at a rate not exceeding 0.5 inch per minute.

4.4.17.3 Tests on the snubber component in the axial direction. The snubber component shall be arranged in a manner similar to that shown on figure 4 and subjected to four loading and unloading cycles in the axial direction, each to 150,000 pounds. The cycles shall be conducted at a rate not exceeding 0.5 inch per minute. The deflections obtained at 50,000-pound load and at 150,000-pound load on the fourth loading cycle shall meet the requirements of 3.3.4.



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4.4.18 Drift test of individual shear and compression mounts. The compression and the shear components previously tested for dynamic stiffness shall be tested individually for drift. Loads of 8,400 and 1,600 pounds shall be applied to the individual compression and shear mount, respectively, in the axial direction. The height of each component shall be measured to the nearest 0.001 inch 1 hour after application of the load and again after 14 days. The difference in the two readings in mils shall be taken as the drift of the mount. At the end of the 14 days, within 15 minutes after the final reading has been obtained, the dynamic stiffness in the axial direction shall be determined while each mount is at rated load. Temperature shall not vary by more than ±2 °F during the drift test.

4.4.19 Porosity and delamination. The compression and the shear components shall be cut into two parts along a vertical plane through the center of the component. The cut surfaces of these components shall be carefully examined for porosity. The parts shall be immersed in technical grade toluene for 24 hours. After removal from the toluene, the rubber elements shall be examined for evidence of separation into distinct layers or laminations.

4.5 Toxicity and prohibited materials. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.2.5.1 and 3.2.5.2, as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the mount using data provided by the manufacturer/distributor to the NMCPHC (see 3.2.5.1 and 6.10).

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mount assembly covered in this specification is intended for noise and vibration attenuation. The mount is primarily for use in naval submarines and, where applicable, on naval surface ships.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether metal parts are to be manufactured from steel or manganese bronze (see 3.2.1).

c. Protective treatment, if different from that specified on NAVSEA Drawing 803-1385873 (see 3.2.3).

d. Packaging requirements (see 5.1).

e. Marking (see 6.8).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 21649 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.



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6.3.1 Additional qualification testing provisions. NAVSEA reserves the right to witness tests and perform any of the tests set forth herein where such testing is deemed necessary to ensure compliance to prescribed requirements of the qualification tests. Qualification testing may be conducted only after NAVSEA has authorized such testing in writing. Copies of SD-6, "Provisions Governing Qualification," are available online at http://quicksearch.dla.mil.

6.3.2 Spring constant values. Spring constant values are to be verified by NAVSEA during qualification (see 6.3.1) and when requested by the mount manufacturer.

6.4 Definitions.

6.4.1 Axial direction. A direction parallel to the bolt axis of the assembled mount. When the mount is in service, the axial direction is normally the vertical direction.

6.4.2 Radial direction. A direction perpendicular to the bolt axis of the assembled mount. When the mount is in service, the radial direction is normally the horizontal direction.

6.5 Mount components. The compression, shear, and snubber components of this mount assembly are not intended to be used separately. All three components are required for shipboard installation.

6.6 Suggested formulations (see 6.7).

6.6.1 Suggested formulations and application procedure for oil-ozone resistant coating.

6.6.1.1 Suggested formulations for coating. The formulations shown below should be blended together just prior to use. The viscosity of the blend should be adjusted to about 650 centipoises by using more or less methyl ethyl ketone in the formulations.

TABLE XV. Suggested coating formulations.

| | **A** | **B** |
|---|---|---|
| Prefluxed blend of 70% nitrile rubber with 30% polyvinyl chloride | 100 | 100 |
| N330 carbon black | 30 | 30 |
| Stearic acid | 1 | 1 |
| Zinc oxide | 5 | 5 |
| CPB (dibutyl vanthogen disulfide) | 8 | --- |
| DBA (dibenzyl amine) | --- | 8 |
| Sulfur | --- | 4 |
| Methyl ethyl ketone | 656 | 674 |

6.6.1.2 Suggested procedure for application of coating to natural rubber and BUNA-N components. The sequence of operations shown below should be followed in order to obtain satisfactory performance from the coating:

a. Remove rubber flash from component by trimming or buffing.
b. Sandblast the component and brush off dust.
c. Apply primer in accordance with manufacturer's recommendation to all metal surfaces.
d. Wash the rubber portion of the component with an aqueous solution of Vel detergent, or equivalent, and rinse with tap water.
e. Chlorinate the rubber surfaces by immersing each component for 3 minutes in water saturated with chlorine.
f. Rinse mount components with tap water and dry until all traces of water have evaporated.
g. Blend the A solution with the B solution and let stand 1 hour to permit entrapped air to escape.



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h. Apply two coats of the blended formulations to the entire component, including metal parts, by brushing or dipping.

i. Allow the first coat to dry for at least 1 hour and the second coat to dry for at least 4 days before handling the mount; alternatively, the second coat should be allowed to dry for at least 4 hours, and then the component placed in an oven at 130±5 °F for 16 hours to cure the coating.

6.6.2 Suggested formulations of rubber stocks. The following are suggested formulations of rubber stocks:

TABLE XVI. Suggested rubber stock formulations.

| **Shear and compression components** | | **Snubber component** | | **Bonding cement for snubber component** | |
|---|---|---|---|---|---|
| Natural rubber | 100 | Butadiene-high acrylonitrile copolymer | 100 | Two step phenolic resin | 100 |
| N990 carbon black | 15 | N770 carbon black | 75 | Methyl ethyl ketone | 150 |
| Stearic acid | 1 | Stearic acid | 1 | Methanol | 150 |
| Zinc oxide | 5 | Zinc oxide | 5 | | |
| Octylated diphenylamine | 1 | Two step phenolic resin | 15 | | |
| Benzothiazyl disulfide | 1 | Coumarone-indene resin | 30 | | |
| Tetramethyl thiuram monosulfide | 0.3 | Salicylic acid | 0.5 | | |
| Sulfur | 1.5 | Benzothiazyl disulfide | 1.5 | | |
| | | Sulfur | 1.5 | | |
| Total | 124.8 | | 229.5 | | 400 |
| NOTE: Rubber formulations based on parts per hundred rubber (phr). | | | | | |

6.7 Disclaimer. The furnishing of information, formulations, and procedures (see 6.6) are solely for assistance in the fabrication of the 5M10,000-H mount and for naval use. The use of this information does not constitute any agreement or obligation by the Department of the Navy to acquire mounts made of these formulas. Also, the use of this information does not guarantee compliance with this specification, nor will it relieve the contractor from having the mounts tested against the applicable test requirements of this specification. Furthermore, the Government does not guarantee that the mounts made using these formulations will conform to the requirements of this specification.

6.8 Marking. Waterproof marking should be provided on interior packages and exterior shipping containers with the following information (see 6.2):

MOUNT TYPE:

DEFENSE SPEC NO: MIL-DTL-21649

DATE OF MFG:

LOT NO:

SHELF LIFE: 7 YEARS FROM DATE OF MFG

Note: The date of manufacture should correspond to the elastomer cure date.



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6.9 Measurement of test apparatus stiffness. Where practicable, the stiffness of those components of the test apparatus in series with the specimen should be determined experimentally as follows:

a. For test machines capable of such operation, place the machine in the load feedback mode with the displacement transducer on its most sensitive range and the force transducer on its full scale setting.

b. Bring the platens of the machine either directly in contact or against a stiff steel spacer separating the platens. **Caution: This procedure is not recommended for test machines not equipped with a safety interlock system to prevent the force from exceeding a predetermined value.**

c. The force applied should be sufficient to ensure good resolution from both the force and the displacement instrumentation. Sufficient points should be taken to ensure that the force-deflection relationship is linear and that effects, such as non-parallelism between platens, do not produce nonlinear results.

d. The stiffness of the test apparatus $K_{apparatus}$ should be determined from the "best fit" straight line through the data points of the force-deflection curve, disregarding anomalous (i.e., abnormal or irregular) points, which may occur at low loads prior to proper seating of the test machine components.

e. Where it is not practical to experimentally determine the stiffness of the test apparatus, manufacturer's published stiffness for each component in series with the test specimen should be used to calculate the overall stiffness of the test apparatus.

6.10 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.11 Subject term (key word) listing.

Isolation

Rubber

Shock

Snubber

Vibration

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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MAKE (6) CUTS:
1/4” LONG AND 1/8” DEEP
AT 60 DEGREE INTERVALS.

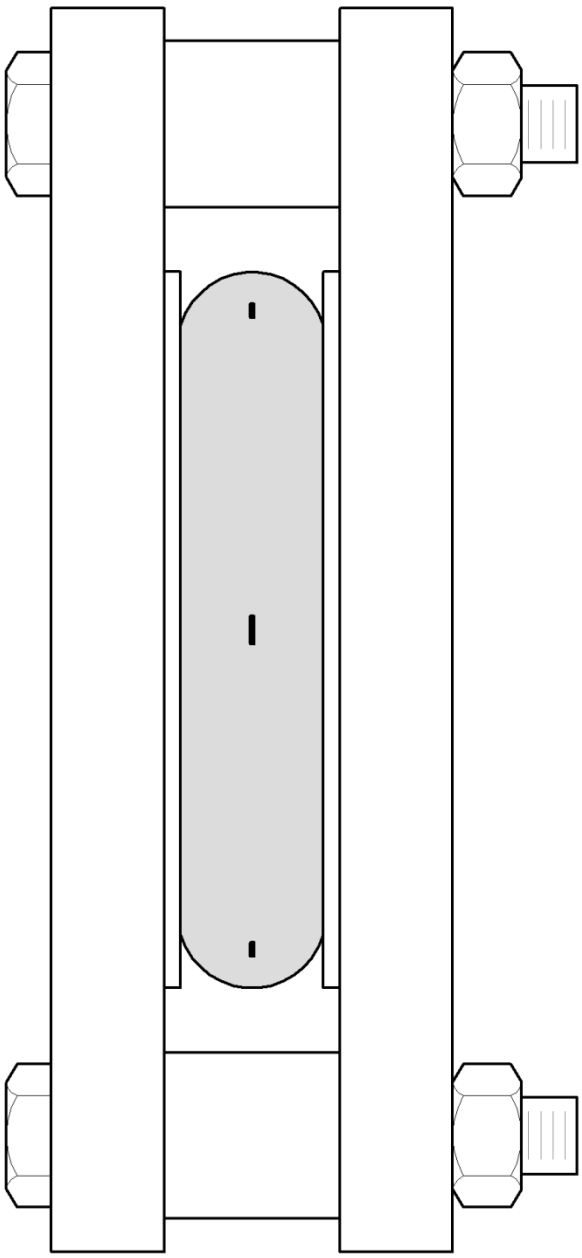


FIGURE 1. Compression sandwich of mount compressed in cut growth test apparatus.



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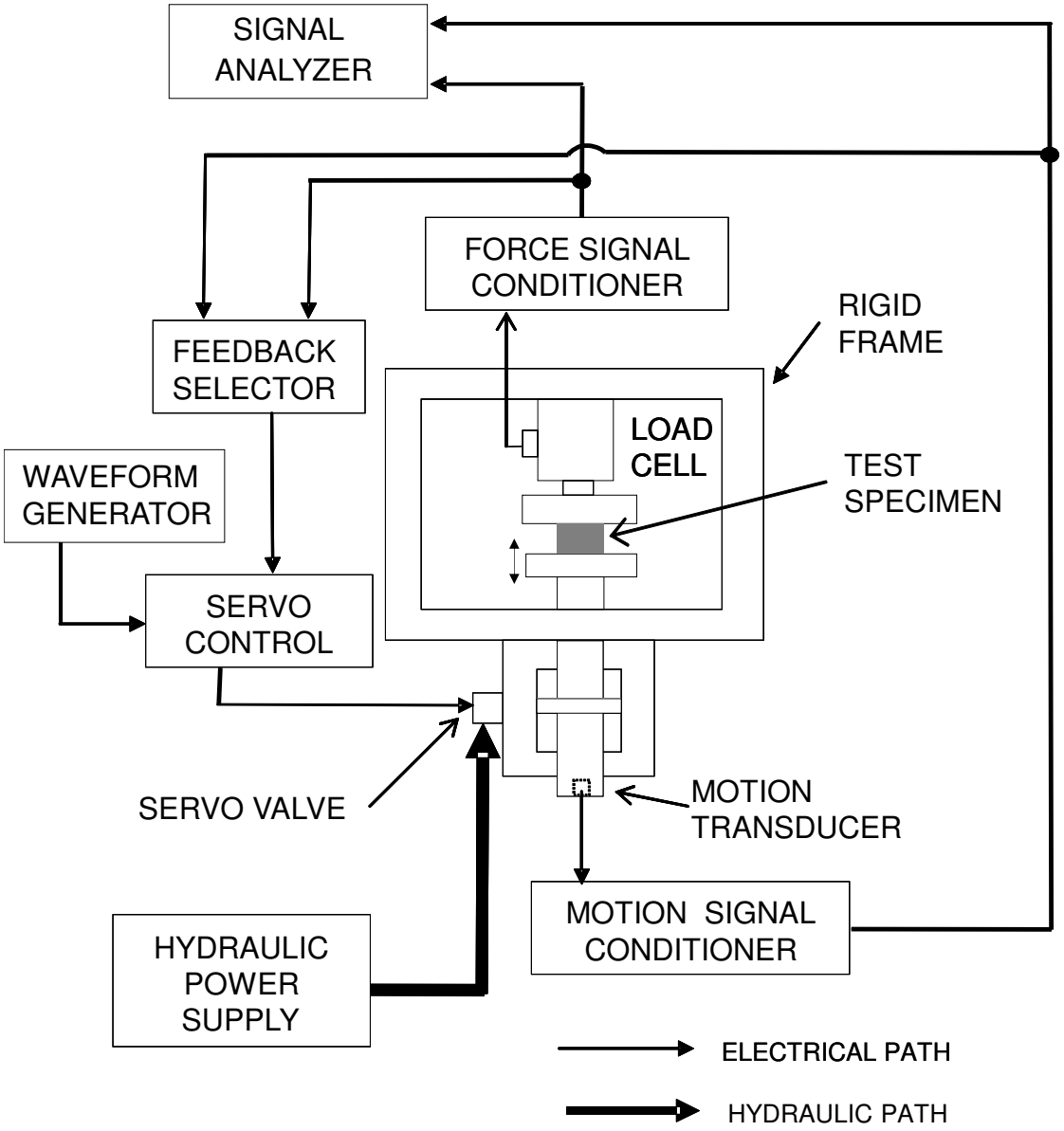



FIGURE 2. Schematic diagram of major components of electrohydraulic closed-loop test machine.



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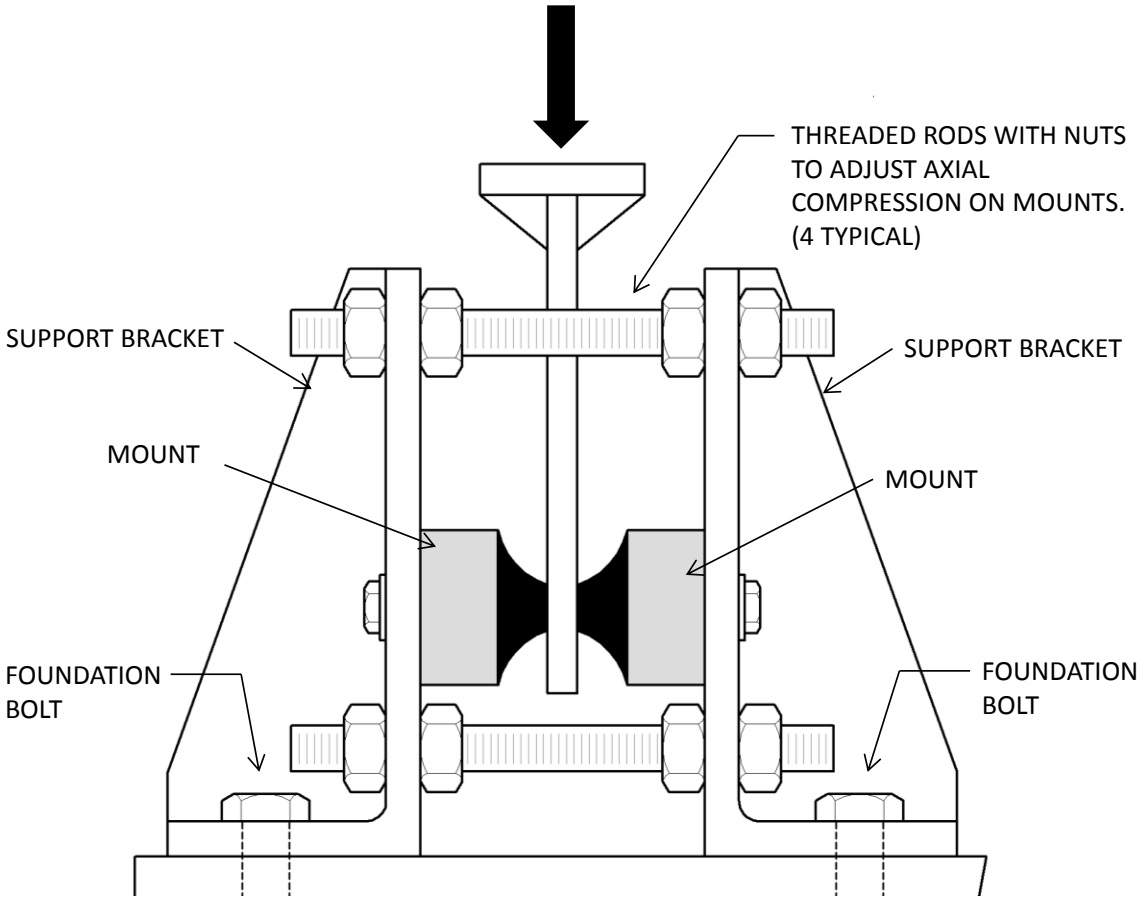



FIGURE 3. Arrangement for conducting load deflection test on shear components in the radial direction.



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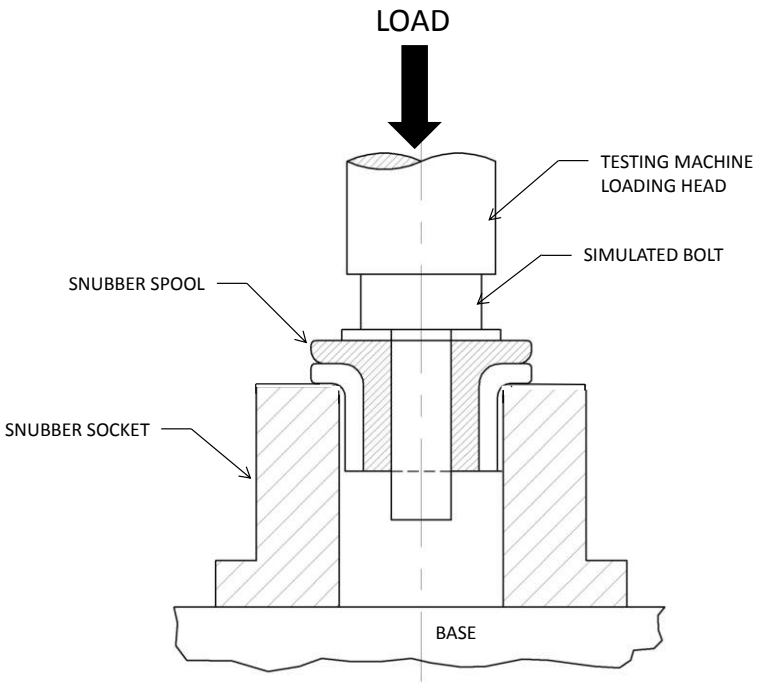



FIGURE 4. Arrangement for conducting static load deflection tests on snubber in the axial direction.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 5342-2016-001)

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